

**Section: 1. PRODUCT AND COMPANY IDENTIFICATION**

Product name : ECO-STAR SOUR

Other means of identification : Not applicable.

Recommended use : Laundry product

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : Product is sold ready to use.

Company : ECOLAB INTERNATIONAL SDN. BHD.  
Suite 12-01, 12-02 & 12-03A, Level 12, The Pinnacle, Persiaran  
Lagoon, Bandar Sunway  
46150 Petaling Jaya, Selangor Malaysia  
603-7628 5200

Emergency telephone number : 1800 181 880

Issuing date : 16.02.2022

**Section: 2. HAZARDS IDENTIFICATION**
**GHS Classification**

Corrosive to metals : Category 1

Skin corrosion/irritation : Category 1B

Serious eye damage/eye irritation : Category 1

**GHS Label element**

Hazard pictograms :



Signal Word : Danger

Hazard Statements : May be corrosive to metals.  
Causes severe skin burns and eye damage.

Precautionary Statements : **Prevention:**  
Keep only in original container. Do not breathe dusts or mists. Wash skin thoroughly after handling. Wear protective gloves/ protective clothing/ eye protection/ face protection.

**Response:**  
IF SWALLOWED: Rinse mouth. Do NOT induce vomiting. IF ON SKIN (or hair): Remove/ Take off immediately all contaminated clothing. Rinse skin with water/ shower. IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing. Immediately call a POISON CENTER or doctor/ physician. IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do so. Continue rinsing. Wash contaminated clothing before reuse. Absorb spillage to prevent material damage.

**Storage:**

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Store locked up. Store in corrosive resistant container with a resistant inner liner.

**Disposal:**

Dispose of contents/ container to an approved waste disposal plant.

**Other hazards** : Do not mix with bleach or other chlorinated products – will cause chlorine gas.

### Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS

Pure substance/mixture : Mixture

| Chemical Name   | CAS-No.   | Concentration: (%) |
|-----------------|-----------|--------------------|
| Phosphoric acid | 7664-38-2 | 30 - 60            |
| citric acid     | 77-92-9   | 5 - 10             |

### Section: 4. FIRST AID MEASURES

In case of eye contact : Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses, if present and easy to do. Continue rinsing. Get medical attention immediately.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention immediately.

If swallowed : Rinse mouth with water. Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

If inhaled : Remove to fresh air. Treat symptomatically. Get medical attention if symptoms occur.

Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

### Section: 5. FIREFIGHTING MEASURES

Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Unsuitable extinguishing media : None known.

Specific hazards during firefighting : Exposure to decomposition products may be a hazard to health.

Hazardous combustion products : Decomposition products may include the following materials:  
Carbon oxides  
Oxides of phosphorus

Special protective equipment : Use personal protective equipment.

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for firefighters

Specific extinguishing methods : Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

#### Section: 6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Avoid inhalation, ingestion and contact with skin and eyes. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.

Environmental precautions : Do not allow contact with soil, surface or ground water.

Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

#### Section: 7. HANDLING AND STORAGE

Advice on safe handling : Do not ingest. Do not breathe dust/fume/gas/mist/vapours/spray. Use only with adequate ventilation. Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing. Do not mix with bleach or other chlorinated products – will cause chlorine gas. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Storage temperature : 0 °C to 50 °C

#### Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

##### Components with workplace control parameters

| Components      | CAS-No.   | Form of exposure | Permissible concentration | Basis  |
|-----------------|-----------|------------------|---------------------------|--------|
| Phosphoric acid | 7664-38-2 | TWA              | 1 mg/m3                   | MY OEL |

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

##### Personal protective equipment

Eye protection : Safety goggles  
Face-shield

Hand protection : Wear the following personal protective equipment:  
Standard glove type.  
butyl-rubber

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Nitrile  
Unsupported neoprene  
Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.

Skin protection : Personal protective equipment comprising: suitable protective gloves, safety goggles and protective clothing

Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
An acid gas cartridge may be used.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use.  
Wash face, hands and any exposed skin thoroughly after handling.  
Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

### Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance : liquid  
Colour : clear, pink  
Odour : odourless  
pH : 0.2 - 1.0, (100 %)  
Flash point : Not applicable., Does not sustain combustion.  
Odour Threshold : no data available  
Melting point/freezing point : no data available  
Initial boiling point and boiling range : > 100 °C  
Evaporation rate : no data available  
Flammability (solid, gas) : Not applicable.  
Upper explosion limit : no data available  
Lower explosion limit : no data available  
Vapour pressure : no data available  
Relative vapour density : no data available  
Relative density : 1.22 - 1.23  
Water solubility : soluble  
Solubility in other solvents : no data available  
Partition coefficient: n-octanol/water : no data available  
Auto-ignition temperature : no data available  
Thermal decomposition : no data available  
Viscosity, kinematic : no data available  
Explosive properties : no data available  
Oxidizing properties : Yes  
Molecular weight : no data available

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VOC : no data available

#### Section: 10. STABILITY AND REACTIVITY

Reactivity : No dangerous reaction known under conditions of normal use.

Chemical stability : Stable under normal conditions.

Possibility of hazardous reactions : Do not mix with bleach or other chlorinated products – will cause chlorine gas.

Conditions to avoid : None known.

Incompatible materials : Metals  
Chlorine-based bleaching agents  
Bases  
Organic materials

Hazardous decomposition products : In case of fire hazardous decomposition products may be produced such as:  
Carbon oxides  
Oxides of phosphorus

#### Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

##### Potential Health Effects

Eyes : Causes serious eye damage.

Skin : Causes severe skin burns.

Ingestion : Causes digestive tract burns.

Inhalation : May cause nose, throat, and lung irritation.

Chronic Exposure : Health injuries are not known or expected under normal use.

##### Experience with human exposure

Eye contact : Redness, Pain, Corrosion

Skin contact : Redness, Pain, Corrosion

Ingestion : Corrosion, Abdominal pain

Inhalation : Respiratory irritation, Cough

##### Toxicity

##### Product

Acute oral toxicity : no data available

Acute inhalation toxicity : no data available

Acute dermal toxicity : no data available

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Skin corrosion/irritation : no data available  
Serious eye damage/eye irritation : no data available  
Respiratory or skin sensitization : no data available  
Carcinogenicity : no data available  
Reproductive effects : no data available  
Germ cell mutagenicity : no data available  
Teratogenicity : no data available  
STOT - single exposure : no data available  
STOT - repeated exposure : no data available  
Aspiration toxicity : no data available

#### Components

Acute oral toxicity : Phosphoric acid  
LD50 rat: > 2,600 mg/kg  
  
citric acid  
LD50 rat: 11,700 mg/kg

#### Components

Acute inhalation toxicity : Phosphoric acid  
4 h LC50 rat: 0.962 mg/l  
Test atmosphere: dust/mist

#### Components

Acute dermal toxicity : Phosphoric acid  
LD50 rabbit: > 2,000 mg/kg

### Section: 12. ECOLOGICAL INFORMATION

#### Toxicity

Environmental Effects : This product has no known ecotoxicological effects.

#### Product

Toxicity to fish : no data available  
Toxicity to daphnia and other aquatic invertebrates : no data available  
Toxicity to algae : no data available

#### Components

Toxicity to fish : citric acid  
96 h LC50 Fish: > 100 mg/l

#### Components

Toxicity to daphnia and other aquatic invertebrates : Phosphoric acid  
48 h EC50 Daphnia magna (Water flea): > 100 mg/l

#### Components

Toxicity to algae : Phosphoric acid  
72 h EC50 Desmodesmus subspicatus (green algae): > 100 mg/l

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#### Persistence and degradability

Poorly biodegradable

#### Bioaccumulative potential

no data available

#### Mobility in soil

no data available

#### Other adverse effects

no data available

### Section: 13. DISPOSAL CONSIDERATIONS

Disposal methods : Do not contaminate storm water drains, natural waterways or soil with chemical or used container. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of contents/container in accordance with local regulations Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

### Section: 14. TRANSPORT INFORMATION

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

#### Land transport

UN number : 1805  
Proper shipping name : PHOSPHORIC ACID, SOLUTION  
Class : 8  
Packing group : III  
Hazchem Code : 2R  
Environmentally hazardous : No

#### Sea transport (IMDG/IMO)

UN number : 1805  
Proper shipping name : PHOSPHORIC ACID SOLUTION  
Class : 8  
Packing group : III  
Marine pollutant : No

### Section: 15. REGULATORY INFORMATION

#### National Regulations

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulation 2013  
Occupational Safety and Health Act 1994  
Environmental Quality Act 1974

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The components of this product are reported in the following inventories:

**United States TSCA Inventory :**

All substances listed as active on the TSCA inventory

**Canadian Domestic Substances List (DSL) :**

This product contains one or several components that are not on the Canadian DSL nor NDSL.

**Australia. Australian Industrial Chemicals Introduction Scheme (AICIS) :**

not determined

**New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :**

not determined

**Japan. ENCS - Existing and New Chemical Substances Inventory :**

not determined

**Korea. Korean Existing Chemicals Inventory (KECI) :**

On the inventory, or in compliance with the inventory

**Philippines Inventory of Chemicals and Chemical Substances (PICCS) :**

On the inventory, or in compliance with the inventory

**China Inventory of Existing Chemical Substances :**

On the inventory, or in compliance with the inventory

**Taiwan Chemical Substance Inventory :**

not determined

### Section: 16. OTHER INFORMATION

|                     |                      |
|---------------------|----------------------|
| Issuing date        | : 16.02.2022         |
| Date of first issue | : 22.02.2015         |
| version             | : 1.3                |
| Prepared by         | : Regulatory Affairs |

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.